



Toilet Seva

Compound Engineering — Final Plan v2.0

July 2026 | PWA + Play Store

Strategy & Vision

Vision: अहिल्यानगरातील प्रत्येक नागरिकाला कुठेही असताना जवळचे स्वच्छ शौचालय लगेच सापडेल — आणि त्याची गुणवत्ता सुधारण्यात त्यांचा सहभाग असेल.

Goal: Ahilyanagar Mahanagar Palika चा Swachh Survekshan rank सुधारणे + नागरिकांचा विश्वास वाढवणे.

Unique Position: India मधील पहिला palika-owned, Marathi-native, live-tracked public toilet PWA.

10 Unique Features (ce-ideate)



1. Live GPS Tracking

watchPosition वापरून list आपोआप update — तुम्ही चालत असताना जवळचे टॉयलेट बदलत राहते



2. Smart Filter + Search

महिला | दिव्यांग | स्वच्छ ★★★★★ — filter chips + real-time search bar



3. Photo Complaint Camera थेट उघडतो — photo attach करून तक्रार नोंदवा, admin ला proof मिळतो	4. Voice Complaint (Marathi) Mic button — Marathi मध्ये बोला, text आपोआप भरतो (Web Speech API)
5. WhatsApp Share One tap → WhatsApp वर toilet location share करा — organic spread	6. Dark Mode Auto device detection — रात्री वापरायला सोपं, AMOLED battery save
7. Citizen Points Rating +5 Complaint +10 Resolved +15 — gamification for engagement	8. Admin Stats Dashboard Total Open Avg Rating Resolved — live stats + top complaints chart
9. Browser Notifications Complaint resolve झाल्यावर citizen ला push notification	10. Cluster Map Red/Green markers + live blue dot — zoom out वर cluster, zoom in वर individual

Project Structure

File	काय आहे
index.html	Main app — सर्व 4 screens एकत्र
manifest.json	PWA config — name, icon, theme color
service-worker.js	Offline cache — internet नसेल तरी चालतं
logo.png	Mahanagar Palika official logo
icon.svg	App icon — home screen वर दिसतो

4 App Screens

Screen 1: यादी (Home)

Header: Logo + टॉयलेट सेवा + Points
Location: Live GPS dot — real-time tracking

Search: "नाव किंवा परिसर शोधा..."

Filters: सर्व महिला दिव्यांग उपलब्ध ★★★★★

Cards: Name | Distance | Rating | Type | Facilities | Navigate | Rate | Share

Screen 2: नकाशा (Map)

Leaflet.js + OpenStreetMap (free, no API key)

Open toilets | Closed toilets | Your location (live)

Click marker → Popup with name + Navigate button

Screen 3: तक्रार (Complaint)

Toilet निवडा | Type निवडा | Voice text | Photo | Submit

Success → +10 points earned | Browser notification request

Screen 4: Admin Dashboard

Password: admin123

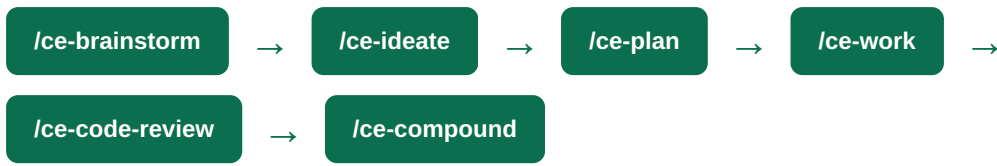
Stats: Total | Open Complaints | Avg Rating | Resolved Today

Tabs: Complaints (with photos) | Toilet list | Add new toilet

Tech Stack

Feature	Technology	Cost
Frontend	HTML5 + CSS3 + Vanilla JS	Free
Maps	Leaflet.js + OpenStreetMap	Free
GPS	Web Geolocation API (watchPosition)	Free
Voice	Web Speech API — Marathi (mr-IN)	Free
Photo	input capture="environment"	Free
Offline	Service Worker (cache-first)	Free
Notifications	Web Notifications API	Free
Dark Mode	CSS prefers-color-scheme	Free
Storage	localStorage	Free
Hosting	Netlify	Free
Play Store	PWABuilder + Google Play Console	\$25 एकदाच

□ Compound Engineering Workflow



□ Claude Code Prompt

हा prompt Claude Code terminal मध्ये paste करा:

```
Read PLAN_v2.md carefully and build the complete Toilet Seva PWA v2.0.

Create index.html with ALL features:
- Mahanagar Palika logo.png in header + citizen points badge
- Live GPS tracking with watchPosition (auto-update on move)
- Search bar + filter chips (सर्व/महिला/दिव्यांग/उपलब्ध/****)
- WhatsApp share button on each toilet card
- Dark mode (prefers-color-scheme)
- Citizen points (+5 rating, +10 complaint, +15 resolved)
- Voice complaint – Web Speech API, lang: mr-IN
- Photo attachment in complaint form
- Admin stats dashboard (4 cards: Total/Open/Rating/Resolved)
- Browser notifications for complaint status updates
- 10 sample toilets (Ahilyanagar locations)
- All UI text in Marathi

Create manifest.json, service-worker.js (cache logo.png too).

Use /ce-work to execute, then /ce-simplify-code, then /ce-code-review.
```

□ After Build Checklist

- ☐ `npx serve .` — local test करा
- ☐ GPS tracking — phone वर test करा
- ☐ All 4 screens navigate correctly
- ☐ Filter chips work
- ☐ WhatsApp share opens
- ☐ Admin login (admin123) works
- ☐ Dark mode auto-switches
- ☐ Voice complaint records Marathi
- ☐ Photo upload previews correctly
- ☐ Netlify deploy — drag and drop folder
- ☐ PWABuilder — APK download
- ☐ Play Store — submit APK (\$25)

Deployment Path

VS Code



GitHub



Netlify



PWABuilder



Play Store

□Toilet Seva v2.0 — Ahilyanagar Mahanagar Palika | Compound Engineering Plan | July 2026
निरंतर नगरसेवा